UNIT 3: LINEAR TRANSFORMATIONS AND INNER PRODUCT SPACES

Chapter 42: Matrix and linear transformations

Definition: Let V and W be vector spaces. A function $T: V \to W$ is called a **linear transformation** of V into W if

(a) $T(\mathbf{u}+\mathbf{v}) = T(\mathbf{u}) + T(\mathbf{v})$ for every $\mathbf{u}, \mathbf{v}$ in V.

(b) $T(c\mathbf{u}) = cT(\mathbf{u})$ for every $\mathbf{u}$ in V and c in$\mathbb{R}$.

V is called the **domain space** and W is called the **codomain space** of T.

For $T: V \to W$, $\mathbf{w}$ is the **image** of a vector $\mathbf{v}$ if $T(\mathbf{v}) = \mathbf{w}$.

A linear transformation $T: V \to V$ is also called a **linear operator** on V.

Example: Consider $T : \mathbb{R}^3 \to \mathbb{R}^2$ by $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_3 \\ x_2 \end{bmatrix}$. Show this is a linear transformation.

Definition: A function $T : \mathbb{R}^n \to \mathbb{R}^m$ defined by multiplication by a matrix A so that $T(\mathbf{x}) = A\mathbf{x}$ is called a **matrix transformation**. In this situation, we write T as T_A.

Example: Is $T\left(\begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix}\right) = \begin{bmatrix} x_1 + x_3 \\ x_2 \end{bmatrix}$ a matrix transformation?

Theorem: Every matrix transformation is a linear transformation.
Proof:

Remark: If $T_A : \mathbb{R}^n \to \mathbb{R}^m$ is a matrix transformation (i.e. $T(\mathbf{x}) = A\mathbf{x}$, then A is a ___________ by _________ matrix.

Example: For any vector spaces V and W, the function $Z : V \to W$ by $Z(\mathbf{v}) = \mathbf{0}_W$ is a linear transformation.

Example: For any vector space V, the function $I_V : V \to V$ by $I_V(\mathbf{v}) = \mathbf{v}$ is a linear transformation (called the **identity operator**).

Example: Let $L\colon P_2 \to P_2$ with $L(p(x)) = xp'(x)$. Is L a linear transformation?

Example:
Is $D : M_{22} \to \mathbb{R}$ by $D(A) = \det(A)$ is a linear transformation?

Is $S : M_{22} \to \mathbb{R}$ by $S(A) = \operatorname{tr}(A)$ is a linear transformation?

Theorem: Let $T : V \to W$ be a linear transformation. Then

(a) $T(\mathbf{0}_V) =$_____.

(b) $T(-\mathbf{v}) =$ _______ for $\mathbf{v}$ in V.

(c) $T(r_1\mathbf{v}_1 + r_2\mathbf{v}_2 + \cdots + r_k\mathbf{v}_k) =$ ____________________________________ for all $\mathbf{v}_i$ in V and scalar r_i.

Proof: (a) and (b) (try (c) on your own)

Theorem: Let $T: V \to W$ be a linear transformation. Let $\{\mathbf{v}_1, \mathbf{v}_2, \dots, \mathbf{v}_n\}$ be a basis for V. Then T is uniquely determined by $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \dots, T(\mathbf{v}_n)\}$.
Proof:

Example: Let $T: \mathbb{R}^3 \to \mathbb{R}^2$ be a linear transformation and let $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ be a basis for $\mathbb{R}^3$, where

$$\mathbf{v}_1 = \begin{bmatrix} 0 \\ 3 \\ 0 \end{bmatrix}, \ \mathbf{v}_2 = \begin{bmatrix} 0 \\ 0 \\ -5 \end{bmatrix}, \ \mathbf{v}_3 = \begin{bmatrix} 2 \\ 0 \\ 0 \end{bmatrix}.$$

Suppose that

$$T(\mathbf{v}_1) = \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \ T(\mathbf{v}_2) = \begin{bmatrix} 3 \\ 3 \end{bmatrix}, \ T(\mathbf{v}_3) = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

Find $T\left(\begin{bmatrix} 16 \\ 9 \\ 10 \end{bmatrix}\right)$.

Corollary: Let V and W be vector spaces and let $\{\mathbf{b}_1, \mathbf{b}_2, \ldots, \mathbf{b}_n\}$ be a basis of V. Given any vectors $\mathbf{w}_1, \mathbf{w}_2, ..., \mathbf{w}_n$ in W (they need not be distinct), there exists a unique linear transformation $T : V \to W$ satisfying $T(\mathbf{b}_i) = \mathbf{w}_i$ for each $i = 1, 2, \ldots, n$. In fact, the action of T is as follows: Given $\mathbf{v} = r_1\mathbf{b}_1 + r_2\mathbf{b}_2 + \cdots + r_n\mathbf{b}_n$ in V, r_i in $\mathbb{R}$, then

$$T(\mathbf{v}) = T(r_1\mathbf{b}_1 + r_2\mathbf{b}_2 + \cdots + r_n\mathbf{b}_n) = \rule{5cm}{0.4pt}.$$

Corollary: If $T : \mathbb{R}^n \to \mathbb{R}^m$ is a linear transformation, then there exists a unique $m \times n$ matrix A so that $T = T_A$, where $T_A : \mathbb{R}^n \to \mathbb{R}^m$ is defined by $T_A(\mathbf{x}) = A\mathbf{x}$.
In this case, $T = T_A$ is the matrix transformation induced by the unique matrix A, given in terms of its columns by

$$A = \begin{bmatrix} T(\mathbf{e}_1) & T(\mathbf{e}_2) & \cdots & T(\mathbf{e}_n) \end{bmatrix},$$

where $\mathbf{e}_1, \mathbf{e}_2, \ldots \mathbf{e}_n$ are the standard basis vectors of $\mathbb{R}^n$.

A is called the **standard matrix of** T.

Example: Suppose I want to build a linear transformation T in $\mathbb{R}^2$ which is (counter-clockwise) rotation by $90°$.

Chapter 43: Kernel and Image

Definition: Let $T : V \to W$ be a linear transformation.
The **kernel** of T, denoted by $\ker T$, is the subset of V consisting of all elements $\mathbf{v}$ of V such that $T(\mathbf{v}) = \mathbf{0}_W$.

The **image** of T denoted by $\operatorname{im} T$, consists of all vectors in W that are images under T of vectors in V. Thus $\mathbf{w}$ is in $\operatorname{im} T$ if there is a $\mathbf{v}$ in V such that $T(\mathbf{v}) = \mathbf{w}$.

Remark: Some books refer to image as “range.”

In set notation:

$\ker T =$

$\operatorname{im} T =$

Fact: $\ker T$ is never an empty set, because...

$\operatorname{im} T$ is never an empty set, because...

Theorem: Let $T : V \to W$ be a linear transformation

1. $\ker T$ is a subspace of ______.
2. $\operatorname{im} T$ is a subspace of _______.

Proof: (Pick one)

Definition: Given a linear transformation $T : V \to W$:

$\dim(\ker T)$ is called the **nullity** of T and is denoted $\text{nullity}(T)$.

$\dim(\text{im}\, T)$ is called the **rank** of T and is denoted $\text{rank}(T)$.

Example: Let $T_A : \mathbb{R}^n \to \mathbb{R}^m$ be the linear transformation induced by the $m \times n$ matrix A, i.e.,

$$T_A(\mathbf{x}) = \underline{\hspace{3cm}}$$

Then $\ker T_A =$________________ and $\text{im}\, T_A =$ ________________.

Write $A = \begin{bmatrix} \mathbf{c}_1 & \mathbf{c}_2 & \cdots & \mathbf{c}_n \end{bmatrix}$. Write out $\text{im}\, T_A$ in terms of the $\mathbf{c}_i$.

This means: $\text{im}\, T_A = \text{col} A$, so $\text{rank}(T_A) =$____________

Example: Let $L\colon P_2 \to \mathbb{R}$ be the function defined by

$$L(ax^2 + bx + c) = \int_0^1 \left(ax^2 + bx + c\right) \mathrm{d}x.$$

(a) Why is L a linear transformation?

(b) Find $\ker L$.

(c) Find $\dim \ker L = \text{nullity}(L)$.

Theorem: Let A be an $m \times n$ matrix, and let $T_A : \mathbb{R}^n \to \mathbb{R}^m$ be the linear transformation induced by A, that is $T_A(\mathbf{x}) = A\mathbf{x}$ for all $\mathbf{x}$ in $\mathbb{R}^n$.
What if $\text{rank} A = r$? What can we say about the relationship between $\dim \text{im}\, T_A$ and $\dim \ker T_A$?

It turns out that this result hold in general for all linear transformations of finite-dimensional vector spaces:

Theorem: (Dimension Theorem)
If $T : V \to W$ is a linear transformation between finite-dimensional vectors spaces, then

$$\dim \ker T + \dim \text{im}\, T = \dim(V).$$

In other words,

Example: Apply this theorem to $Z : V \to W$ by $Z(\mathbf{v}) = \mathbf{0}_W$.

Example: Go back to $L\colon P_2 \to \mathbb{R}$ by $L(ax^2+bx+c) = \int_0^1 \left(ax^2+bx+c\right) \mathrm{d}x$ and apply the Dimension Theorem.

Example: Consider $T : M_{2\times 2} \to M_{2\times 2}$ by $T(A) = A - A^T$. You can verify that this is a linear transformation.

(a) Find a basis for the kernel of T.

(b) What's the dimension of the image of T?

(c) Find a basis for the image of T.

Chapter 44: Isomorphisms and Composition

One-to-One and Onto Transformations

Let $T : V \to W$ be a linear transformation.

- T is said to be **onto** if $\operatorname{im} T = W$ (i.e. $\dim \operatorname{im} T = \dim W$).
- T is said to be **one-to-one** if $\mathbf{v}_1 \neq \mathbf{v}_2$ implies that $T(\mathbf{v}_1) \neq T(\mathbf{v}_2)$. An equivalent statement is that T is one-to-one if $T(\mathbf{v}_1) = T(\mathbf{v}_2)$ implies that $\mathbf{v}_1 = \mathbf{v}_2$.

Remark: Some books use the word "surjective" instead of onto, and "injective" instead of one-to-one.

Example: Let $T : \mathbb{R}^3 \to \mathbb{R}^2$ be defined by

$$T\left(\begin{bmatrix} u_1 \\ u_2 \\ u_3 \end{bmatrix}\right) = \begin{bmatrix} u_1 + u_3 \\ u_2 \end{bmatrix}.$$

Is T a one-to-one linear transformation?

Is T an onto linear transformation?

Theorem: Let $T: V \to W$ be a linear transformation. Then T is one-to-one if and only if $\ker T = \{\mathbf{0}_V\}$.
Proof:

Corollary: Let $T: V \to W$ be a linear transformation. Then T is one-to-one if and only if $\dim \ker T = \text{nullity}(T) = 0$.

Example: Is $T : M_{23} \to M_{32}$ by $T(A) = A^T$ a one-to-one linear transformation?

Example: Is $S : M_{22} \to \mathbb{R}$ by $S(A) = \text{tr}(A)$ is a one-to-one linear transformation?

Recall: Any linear transformation $T : \mathbb{R}^n \to \mathbb{R}^m$ can be written as a matrix transformation T_A where A is a $m \times n$ matrix.

Theorem: Let A be an $m \times n$ matrix, and let $T_A : \mathbb{R}^n \to \mathbb{R}^m$ be the linear transformation induced by A, that is $T_A(\mathbf{x}) = A\mathbf{x}$ for all columns $\mathbf{x}$ in $\mathbb{R}^n$.

1. T_A is onto if and only if $\operatorname{rank} A =$ ________.
2. T_A is one-to-one if and only if $\operatorname{rank} A =$ ________.

Example: Suppose $T : M_{22} \to \mathbb{R}^3$ is a linear transformation.

Can T be one-to-one?

Can T be onto?

Example: Suppose $T : P_2 \to \mathbb{R}^4$ is a linear transformation.

Can T be one-to-one?

Can T be onto?

Theorem: If $T: V \to W$ is a linear transformation and $\dim V = \dim W$, then

(a) If T is one-to-one, then T is onto.

(b) If T is onto, then T is one-to-one.

Proof: (Hint: Use the Dimension Theorem)

Isomorphisms

What you might be noticing is that vector spaces may be represented differently, but act the same.

Example: How do we add two polynomials $ax+b$ and $cx+d$ in P_1?

How do we add two column vectors $\begin{bmatrix} a \\ b \end{bmatrix}$ and $\begin{bmatrix} c \\ d \end{bmatrix}$ in $\mathbb{R}^2$?

Definition: A linear transformation $T: V \to W$ is called an **isomorphism** if it is both onto and one-to-one. The vector spaces V and W are said to be **isomorphic** if there exists an isomorphism $T: V \to W$, and we write $V \simeq W$ when this is the case.

Remark: Other books may use a different symbol to say “isomorphic”: for example, $V \cong W$.

Example: $T : P_1 \to \mathbb{R}^2$ by $T(ax+b) =$ is an isomorphism.

Warning: This is not the only isomorphism possible! Try $L : P_1 \to \mathbb{R}^2$ by $L(ax+b) =$

Example: Show that M_{22} and $\mathbb{R}^4$ are isomorphic.

Example: Are M_{22} and P_2 isomorphic?

Example: Let V be a vector space with $\dim V = n$. The coordinate representation $C_B : V \to \mathbb{R}^n$ relative to an ordered basis B in V is an isomorphism between V and $\mathbb{R}^n$.

Theorem: If V and W are finite dimensional spaces, the following conditions are equivalent for a linear transformation $T : V \to W$.

1. T is an isomorphism.
2. If $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ is any basis of V, then $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \ldots, T(\mathbf{v}_n)\}$ is a basis of W.
3. There is at least one basis $\{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ of V such that $\{T(\mathbf{v}_1), T(\mathbf{v}_2), \ldots, T(\mathbf{v}_n)\}$ is a basis of W.

Proof: ((1) implies (2)):

Theorem: If V and W are finite dimensional vector spaces, then V and W are isomorphic if and only if $\dim V = \dim W$.
Proof: (if time)

Warning: If V, W have the same dimension, then there must exist an isomorphism between them. That does NOT mean all linear transformations between V and W are isomorphisms.

Example: Consider the linear transformation $T : P_1 \to \mathbb{R}$ by $T(ax+b) = \begin{bmatrix} a+b \\ 0 \end{bmatrix}$. Is this an isomorphism?

Composition and Inverses

Definition: If $T : V \to W$ and $S : W \to U$ are linear transformations, the **composite** $ST : V \to U$ of T and S is defined as $ST(\mathbf{v}) = S(T(\mathbf{v}))$ for all $\mathbf{v}$ in V. The operation of forming the function ST is called **composition**.

Example: If $T : P_1 \to P_1$ by $T(ax+b) = (b+a)x - a$ and $S : P_1 \to P_1$ by $S(ax+b) = bx$, find ST and TS.

Recall: If V is a vector space, the function $I_V : V \to V$ by $I_V(\mathbf{v}) = \mathbf{v}$ is called the **identity operator**.

Fact: I_V is a linear transformation (In fact, it's an ________________).

Theorem: Let $T : V \to W$, $S : W \to U$, and $R : U \to Y$ be linear transformations.

1. The composite ST is again a linear transformation.
2. $TI_V =$_____ and $I_W T =$_____.
3. $(RS)T =$________

Proof: (of (1)):

Theorem: Let V and W be finite dimensional vector spaces. The following conditions are equivalent for a linear transformation $T : V \to W$.

1. T is an isomorphism.
2. There exists a linear transformation $S : W \to V$ such that $ST = I_V$ and $TS = I_W$.

In this case S is also an isomorphism and is uniquely determined by T: If $\mathbf{w}$ in W is $\mathbf{w} = T(\mathbf{v})$, then $S(\mathbf{w}) = \mathbf{v}$. S is called the **inverse** of T and is denoted by T^{-1}.

Example: Consider $S : P_1 \to P_1$ by $S(ax+b) = bx$. Is S an isomorphism? If so, find S^{-1}.

Example: Consider $T : P_1 \to P_1$ by $T(ax+b) = (b+a)x - a$. Is T an isomorphism? If so, find T^{-1}.